Akoizo Original Practice - 2027 Rotation Form 4 - Answer key

Division C | Remote Sensing | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Distinguish an active lidar measurement from a passive optical radiometer measurement.

answer: Lidar transmits light and measures its return; a passive radiometer measures radiation arriving without an emitted probing signal. (2 points).

2. (2 points) Explain how spectral resolution differs from radiometric resolution.

answer: Spectral resolution describes wavelength-band discrimination; radiometric resolution describes discrimination/sensitivity to signal levels. (2 points).

3. (2 points) Apply Wien's law with b=2.90e-3 m K to a 290 K blackbody. Find peak wavelength.

answer: 0.00001 m (2 points).

Numerical target 1e-05; absolute tolerance 2e-08. Accepted units: , m.

4. (2 points) Why is space-based gravimetry relevant to changing ice mass?

answer: Changes in mass distribution affect gravity measurements; gravimetry constrains mass change rather than directly measuring sea-surface height. (2 points).

5. (2 points) Explain why a geostationary observer and a polar-orbiting observer provide different sampling patterns.

answer: A geostationary observer repeatedly views a similar region while a polar orbiter samples changing ground tracks over much of Earth; coverage and revisit must be evaluated separately. (2 points).

6. (2 points) Compute net incoming minus outgoing top-of-atmosphere energy flux.

Synthetic climate-observation worksheet, not actual mission measurements. CERES energy-flux estimates: incoming solar flux 350 W/m^2, reflected solar flux 104 W/m^2 and outgoing longwave flux 238 W/m^2. GRACE mass-change estimates and JASON sea-level estimates are supplied separately in each problem. Use only the named instrument/data descriptions.

answer: 8 W/m^2 (2 points).

Numerical target 8; absolute tolerance 0.016. Accepted units: , W/m^2.

7. (2 points) A JASON teaching series shows 12, 16 and 20 mm at yearly intervals. Find its endpoint slope.

answer: 4 mm/year (2 points).

Numerical target 4; absolute tolerance 0.008. Accepted units: , mm/year.

8. (2 points) A GRACE teaching estimate changes from 0 to -80 gigatonnes over 2 years. Find mean mass-change rate.

answer: -40 gigatonnes/year (2 points).

Numerical target -40; absolute tolerance 0.08. Accepted units: , gigatonnes/year.

9. (2 points) Why can the JASON and GRACE observations not be substituted for one another?

answer: Radar altimetry constrains surface height; gravimetry constrains mass redistribution. Sea level can also change through thermal expansion, so the signals represent different quantities. (2 points).

10. (2 points) An OCO-2 teaching column-CO2 series changes from 404 to 409 ppm. Find the increase.

answer: 5 ppm (2 points).

Numerical target 5; absolute tolerance 0.01. Accepted units: , ppm.

11. (2 points) A MODIS scene is masked as cloud-covered. Explain why treating missing pixels as zero aerosol concentration is wrong.

answer: Missing retrievals are not measurements of zero; cloud screening creates a sampling limitation that must be retained. (2 points).

12. (2 points) A CrIS teaching dataset supplies temperature estimates while an instrument map supplies radiative flux. Explain why temperature and energy flux have different units and meanings.

answer: Temperature describes thermal state in K; radiative flux is energy transferred per area per time in W/m^2. A physical radiation model is needed to relate them. (2 points).

13. (2 points) Explain a climate pathway from increasing atmospheric CO2 to changing outgoing longwave radiation, with surface temperature initially fixed.

answer: Additional infrared absorption/emission changes radiative transfer and can reduce outgoing longwave flux for the stated initially fixed-temperature comparison, creating an energy imbalance. (2 points).

14. (2 points) A supplied ENSO index changes sign. Why should regional rainfall impacts be evaluated from regional data rather than assumed identical everywhere?

answer: ENSO alters circulation and regional precipitation patterns differently by region and season; a single global uniform rainfall response is not justified. (2 points).

15. (2 points) Distinguish a carbon flux from a carbon reservoir in a climate budget.

answer: A reservoir is stored carbon; a flux is a transfer rate between reservoirs. Both magnitudes and time intervals must be specified. (2 points).

16. (2 points) Why can changing vegetation distributions affect both carbon cycling and surface energy exchange?

answer: Vegetation affects carbon uptake/storage, albedo and evapotranspiration; the sign and size depend on location, species and conditions. (2 points).

17. (2 points) Calculate reflected/incoming solar flux as a dimensionless albedo.

Synthetic climate-observation worksheet, not actual mission measurements. CERES energy-flux estimates: incoming solar flux 350 W/m^2, reflected solar flux 104 W/m^2 and outgoing longwave flux 238 W/m^2. GRACE mass-change estimates and JASON sea-level estimates are supplied separately in each problem. Use only the named instrument/data descriptions.

answer: 0.29714285714285715 (2 points).

Numerical target 0.29714286; absolute tolerance 0.00059428571. Accepted units: , .

18. (2 points) For fixed emissivity, a blackbody temperature doubles. By what factor does emitted power per unit area change?

answer: 16 (2 points).

Numerical target 16; absolute tolerance 0.032. Accepted units: , .

19. (2 points) A supplied linear energy anomaly grows from 4 to 16 W/m^2 over 3 years. Find its slope.

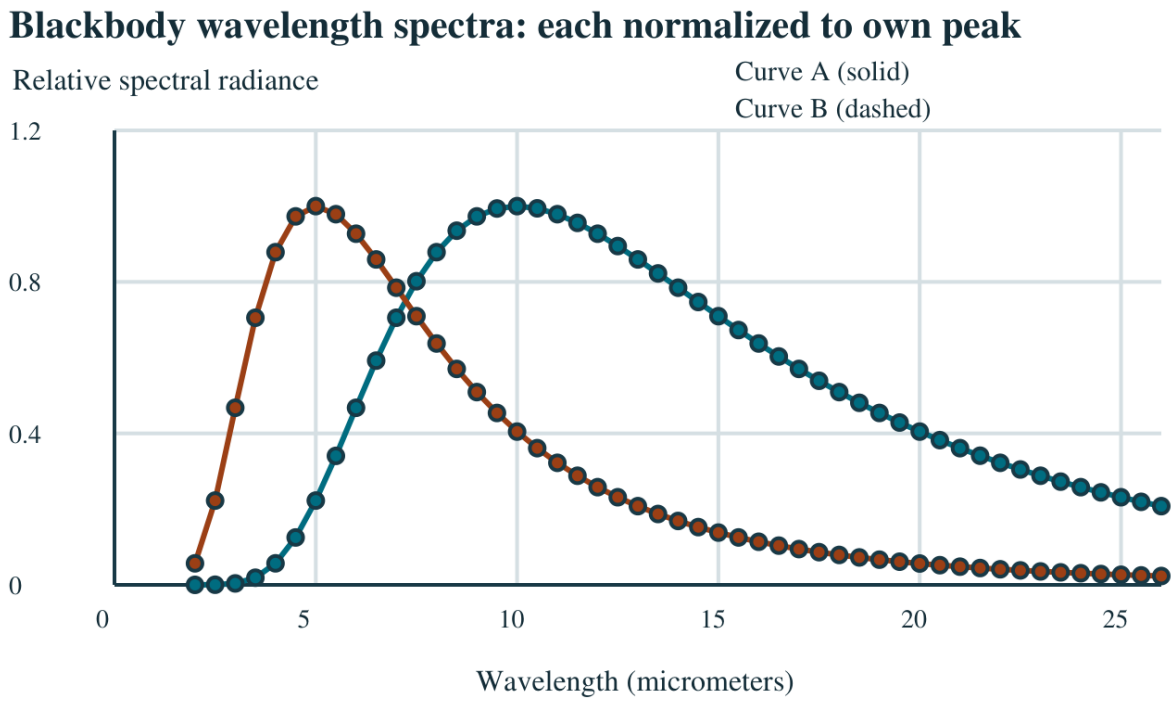
answer: 4 W/m^2/year (2 points).

Numerical target 4; absolute tolerance 0.008. Accepted units: , W/m^2/year.

20. (2 points) Explain the ice-albedo feedback after an initial warming perturbation.

answer: Less reflective ice can expose darker surfaces, increasing absorbed solar energy and reinforcing warming, while acknowledging regional/context dependence. (2 points).

21. (4 points) Use Figure F1. Use the peak of A and b=2900 micrometer K to estimate its temperature.



F1. Normalized thermal spectra (original synthetic teaching figure)

answer: 290 K (4 points).

Numerical target 290; absolute tolerance 0.87. Accepted units: , K.

22. (3 points) Use Figure F1. Which curve represents a higher-temperature blackbody? Explain.

Refer to F1. Normalized thermal spectra (original synthetic teaching figure) shown with the first question in this figure group.

answer: B; its shorter peak wavelength corresponds to higher temperature by Wien law. (3 points).

23. (3 points) Use Figure F1. Why cannot equal plotted peak heights establish equal absolute radiance?

Refer to F1. Normalized thermal spectra (original synthetic teaching figure) shown with the first question in this figure group.

answer: Each curve was normalized separately to its own maximum, removing the absolute-amplitude comparison. (3 points).

24. (5 points) Use Figure F2. Calculate the endpoint JASON sea-level trend.

Synthetic JASON and GRACE observations

Year	JASON sea level (mm)	GRACE ice-mass change (Gt)
0	0	0
1	6	-40
2	12	-80
3	18	-120

F2. Satellite trend panels (original synthetic teaching figure)

answer: 6 mm/year (5 points).

Numerical target 6; absolute tolerance 0.018. Accepted units: , mm/year.

25. (5 points) Use Figure F2. Calculate the endpoint GRACE ice-mass trend.

Refer to F2. Satellite trend panels (original synthetic teaching figure) shown with the first question in this figure group.

answer: -40 Gt/year (5 points).

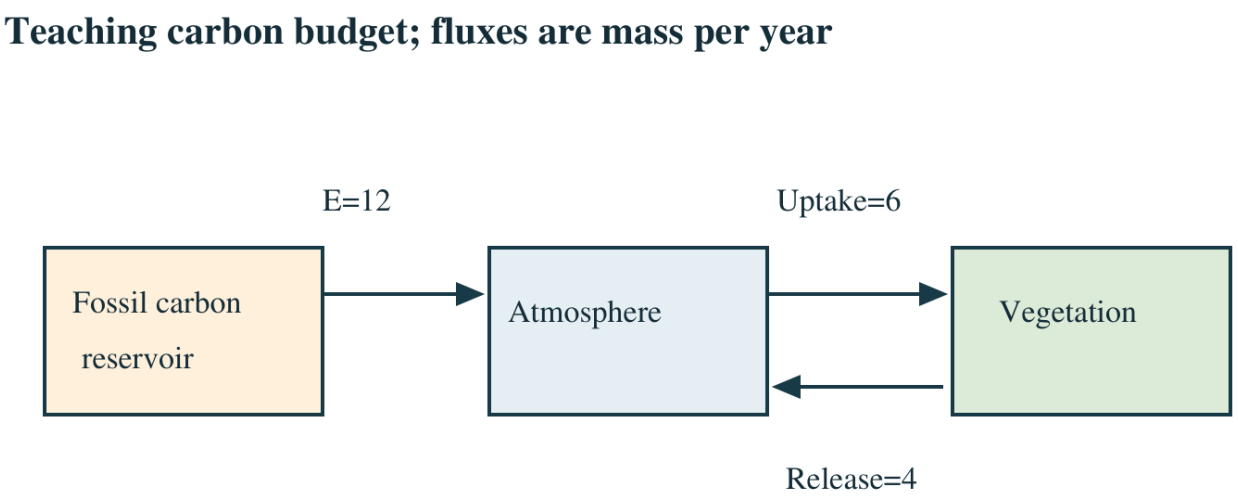
Numerical target -40; absolute tolerance 0.12. Accepted units: , Gt/year.

26. (4 points) Use Figure F2. Why cannot a sea-level trend in mm/year simply replace the GRACE trend?

Refer to F2. Satellite trend panels (original synthetic teaching figure) shown with the first question in this figure group.

answer: They measure different quantities: surface height versus mass redistribution. Thermal expansion and spatial distribution can also affect sea level, so they are not interchangeable observations. (4 points).

27. (3 points) Use Figure F3. Find the net annual vegetation carbon gain from the shown arrows.



F3. Carbon reservoirs and transfers (original synthetic teaching figure)

answer: 2 mass units/year (3 points).

Numerical target 2; absolute tolerance 0.006. Accepted units: , mass units/year.

28. (3 points) Use Figure F3. Find the net atmospheric gain in this simplified budget.

Refer to F3. Carbon reservoirs and transfers (original synthetic teaching figure) shown with the first question in this figure group.

answer: 10 mass units/year (3 points).

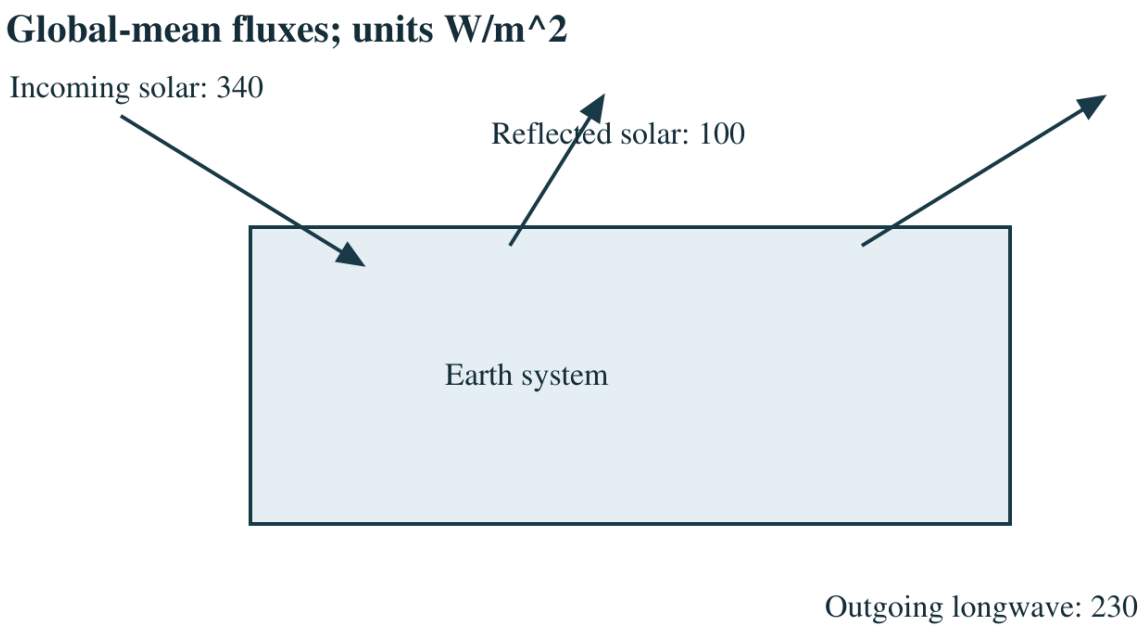
Numerical target 10; absolute tolerance 0.03. Accepted units: , mass units/year.

29. (2 points) Use Figure F3. Distinguish a box quantity from an arrow quantity in this diagram.

Refer to F3. Carbon reservoirs and transfers (original synthetic teaching figure) shown with the first question in this figure group.

answer: A box is a reservoir/storage quantity, while an arrow is a transfer rate between reservoirs; a flux is not the total carbon stock. (2 points).

30. (3 points) Use Figure F4. Calculate the net incoming energy flux.



F4. Top-of-atmosphere energy budget (original synthetic teaching figure)

answer: 10 W/m^2 (3 points).

Numerical target 10; absolute tolerance 0.03. Accepted units: , W/m^2.

31. (3 points) Use Figure F4. Calculate the implied planetary albedo.

Refer to F4. Top-of-atmosphere energy budget (original synthetic teaching figure) shown with the first question in this figure group.

answer: 0.29411765 (3 points).

Numerical target 0.29411765; absolute tolerance 0.00088235294. Accepted units: .

32. (2 points) Use Figure F4. If outgoing longwave and incoming solar initially remain fixed while reflected solar decreases, describe the immediate change in net heating.

Refer to F4. Top-of-atmosphere energy budget (original synthetic teaching figure) shown with the first question in this figure group.

answer: Less reflected solar means more absorbed energy; net incoming energy and immediate heating increase under the stated fixed-flux assumptions. (2 points).